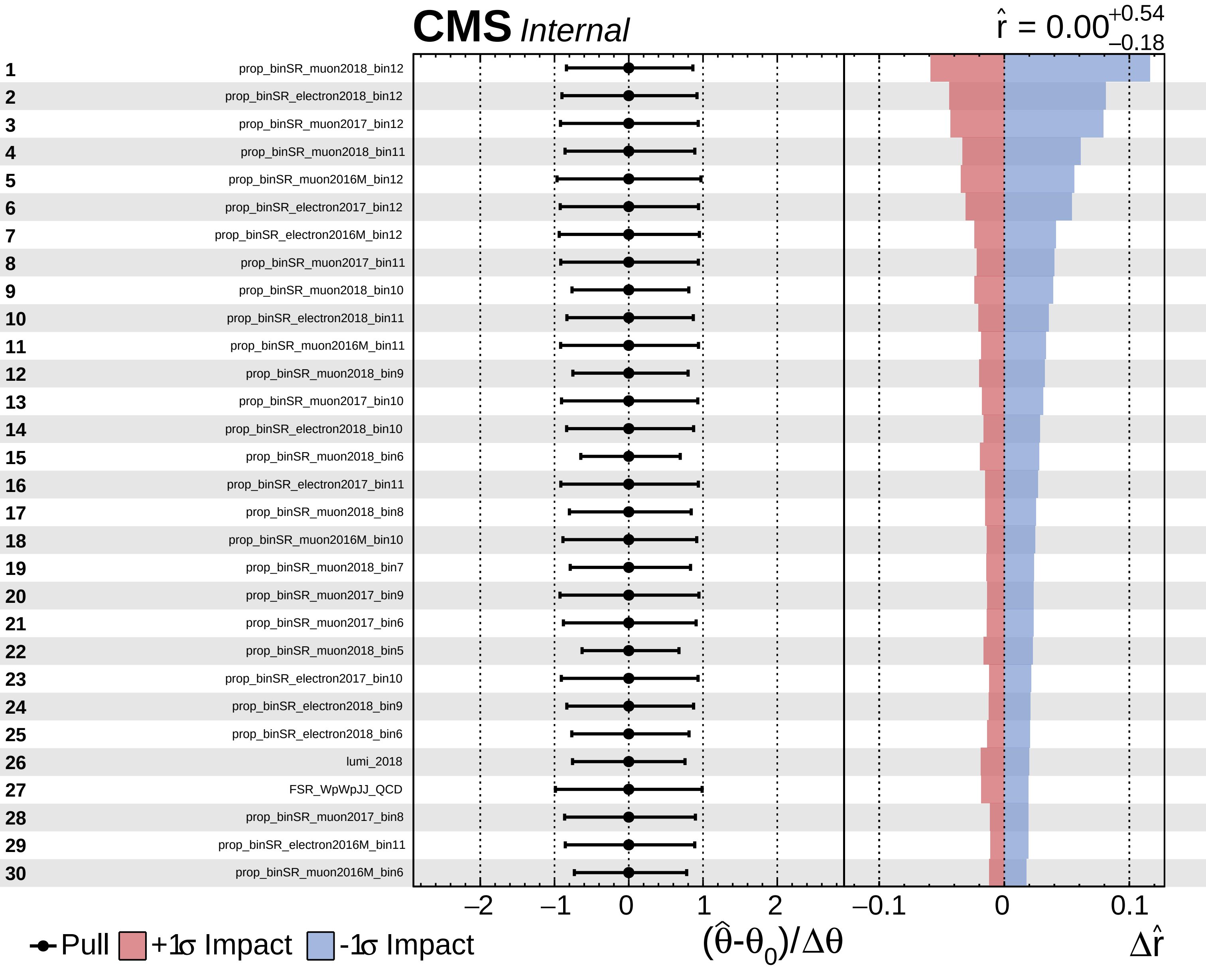
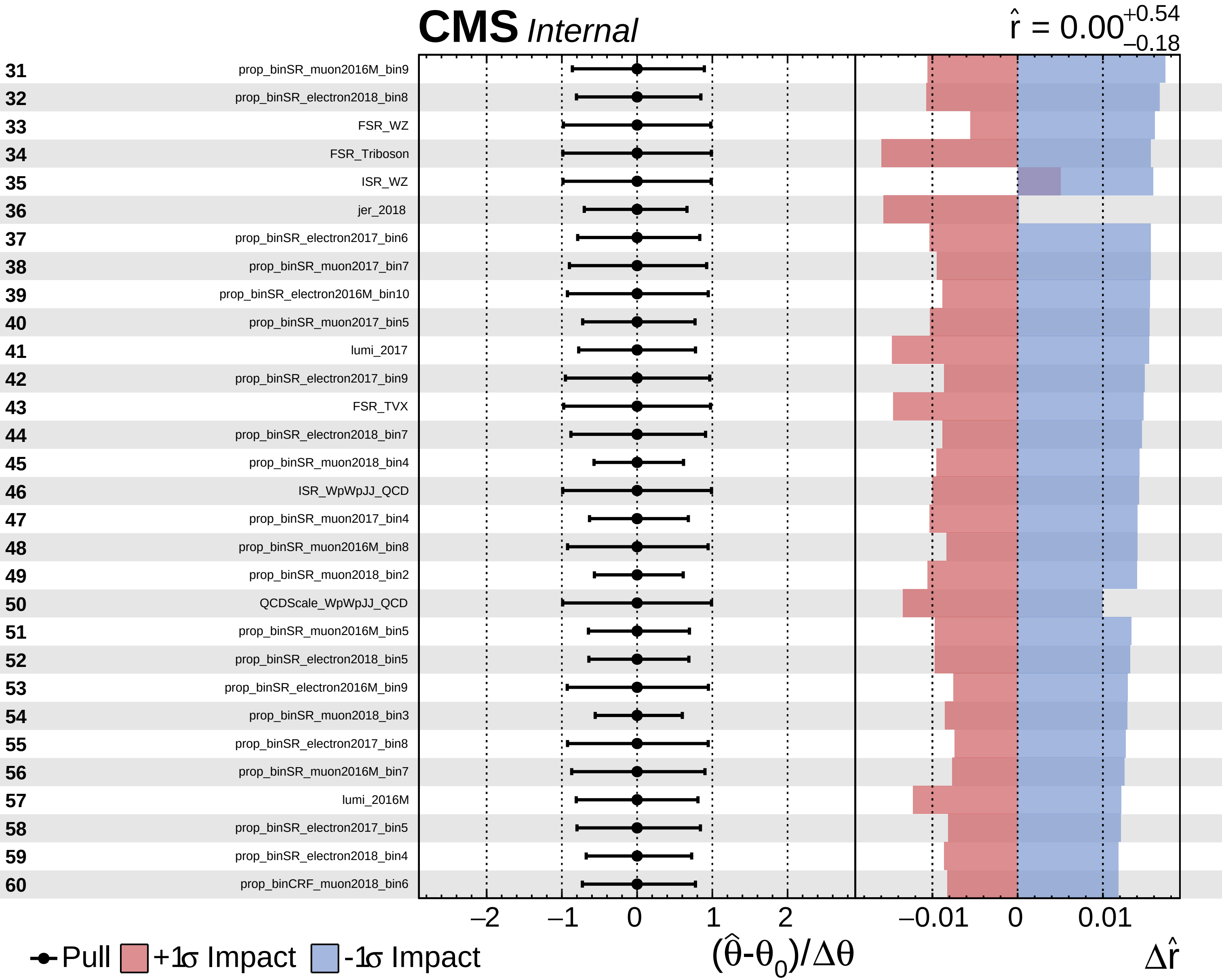
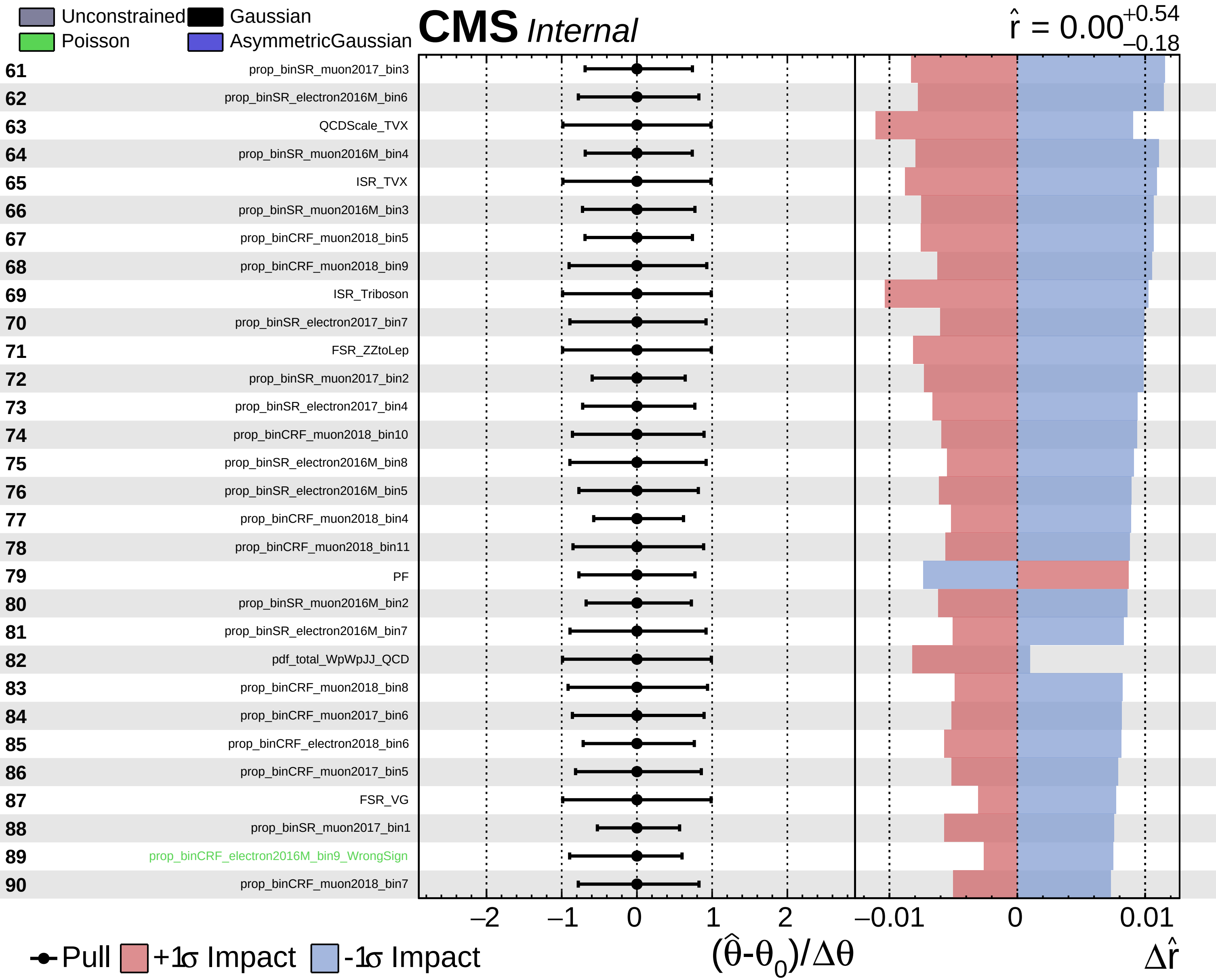
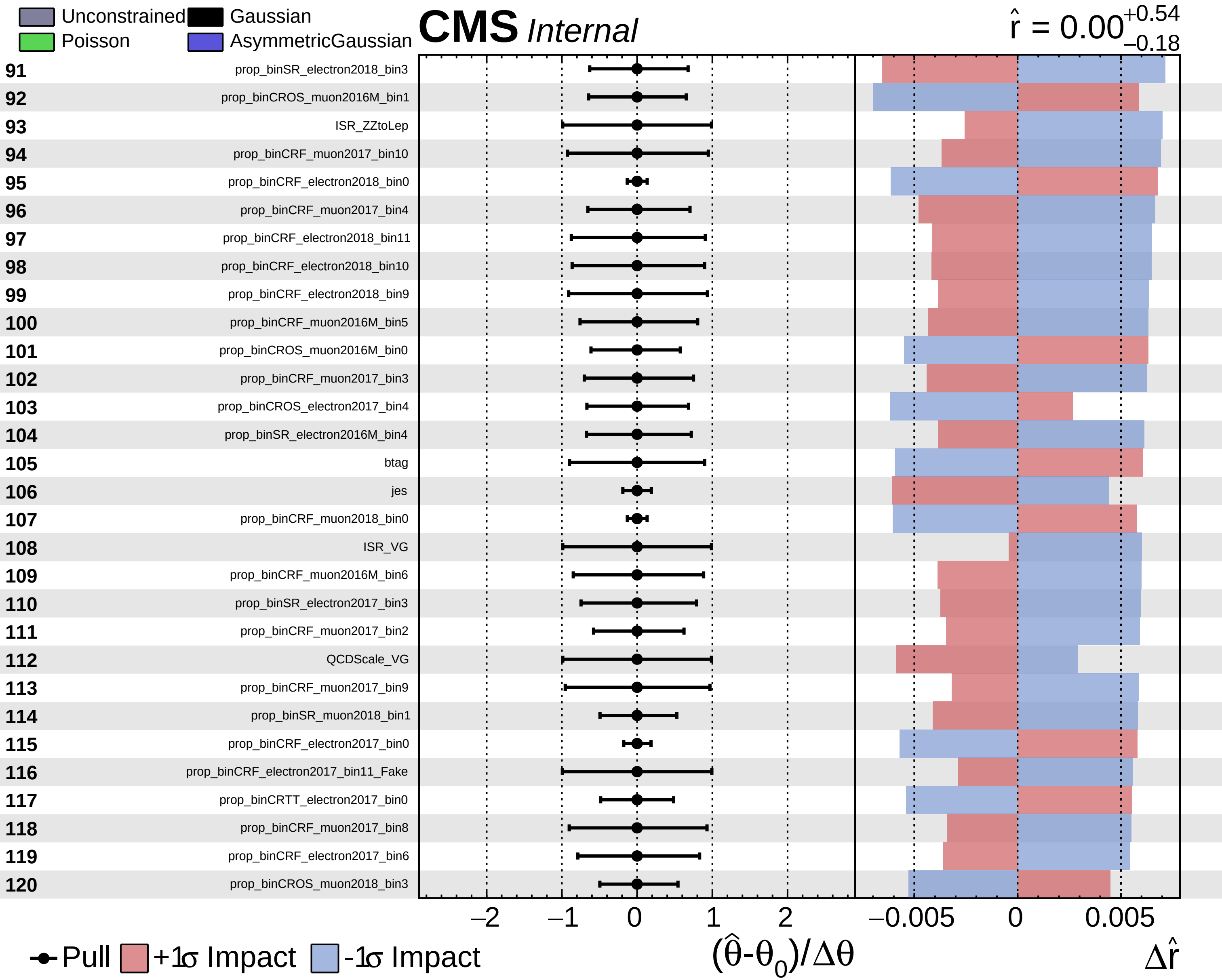

CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
1 prop_binSR_muon2018_bin12
2 prop_binSR_electron2018_bin12
3 prop_binSR_muon2017_bin12
4 prop_binSR_muon2018_bin11
5 prop_binSR_muon2016M_bin12
6 prop_binSR_electron2017_bin12
7 prop_binSR_electron2016M_bin12
8 prop_binSR_muon2017_bin11
9 prop_binSR_muon2018_bin10
10 prop_binSR_electron2018_bin11
11 prop_binSR_muon2016M_bin11
12 prop_binSR_muon2018_bin9
13 prop_binSR_muon2017_bin10
14 prop_binSR_electron2018_bin10
15 prop_binSR_muon2018_bin6
16 prop_binSR_electron2017_bin11
17 prop_binSR_muon2018_bin8
18 prop_binSR_muon2016M_bin10
19 prop_binSR_muon2018_bin7
20 prop_binSR_muon2017_bin9
21 prop_binSR_muon2017_bin6
22 prop_binSR_muon2018_bin5
23 prop_binSR_electron2017_bin10
24 prop_binSR_electron2018_bin9
25 prop_binSR_electron2018_bin6
26 lumi_2018
27 FSR_WpWpJJ_QCD
28 prop_binSR_muon2017_bin8
29 prop_binSR_electron2016M_bin11
30 prop_binSR_muon2016M_bin6
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



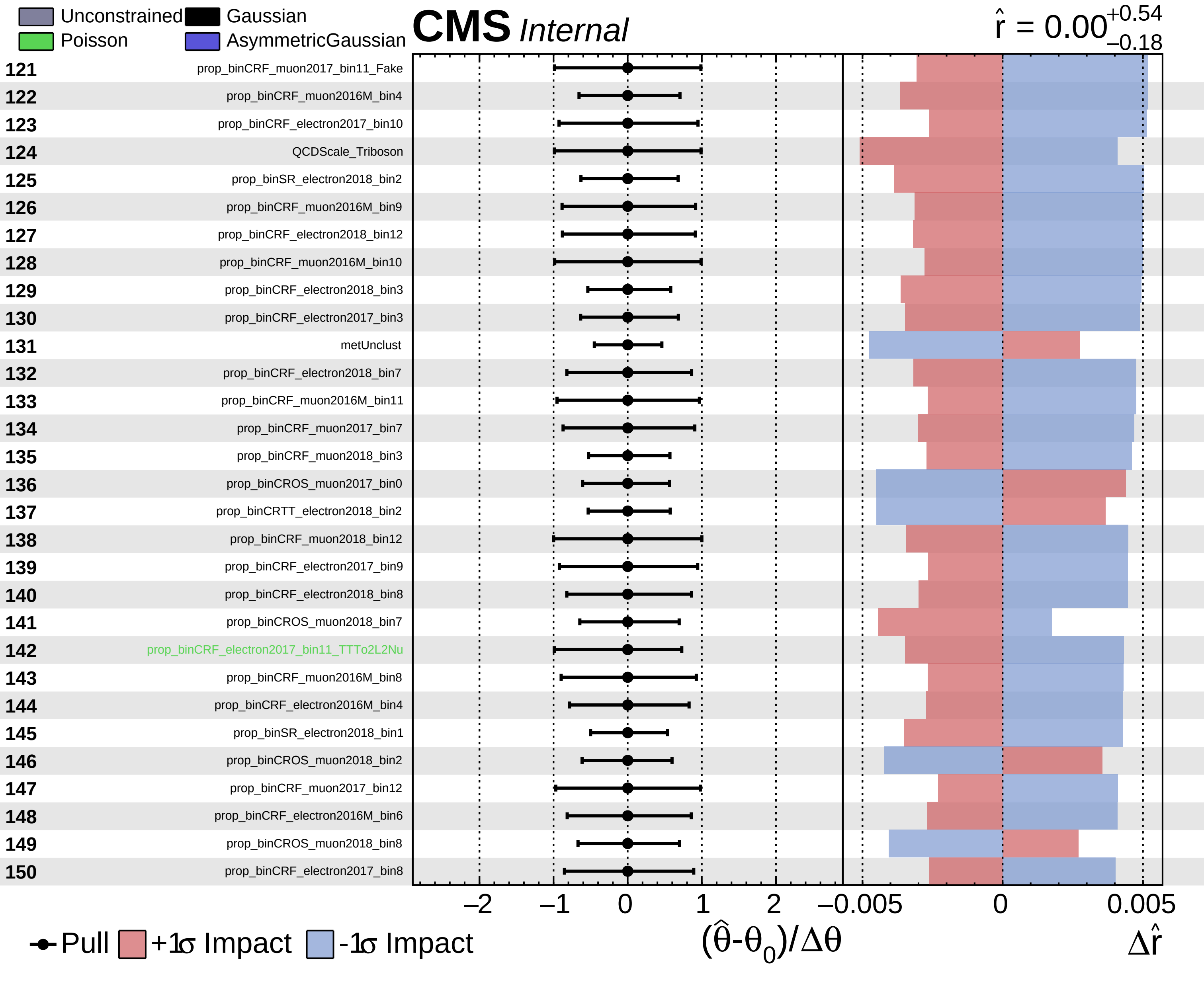
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
31 prop_binSR_muon2016M_bin9
32 prop_binSR_electron2018_bin8
33 FSR_WZ
34 FSR_Triboson
35 ISR_WZ
36 jer_2018
37 prop_binSR_electron2017_bin6
38 prop_binSR_muon2017_bin7
39 prop_binSR_electron2016M_bin10
40 prop_binSR_muon2017_bin5
41 lumi_2017
42 prop_binSR_electron2017_bin9
43 FSR_TVX
44 prop_binSR_electron2018_bin7
45 prop_binSR_muon2018_bin4
46 ISR_WpWpJJ_QCD
47 prop_binSR_muon2017_bin4
48 prop_binSR_muon2016M_bin8
49 prop_binSR_muon2018_bin2
50 QCDScale_WpWpJJ_QCD
51 prop_binSR_muon2016M_bin5
52 prop_binSR_electron2018_bin5
53 prop_binSR_electron2016M_bin9
54 prop_binSR_muon2018_bin3
55 prop_binSR_electron2017_bin8
56 prop_binSR_muon2016M_bin7
57 lumi_2016M
58 prop_binSR_electron2017_bin5
59 prop_binSR_electron2018_bin4
60 prop_binCRF_muon2018_bin6
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



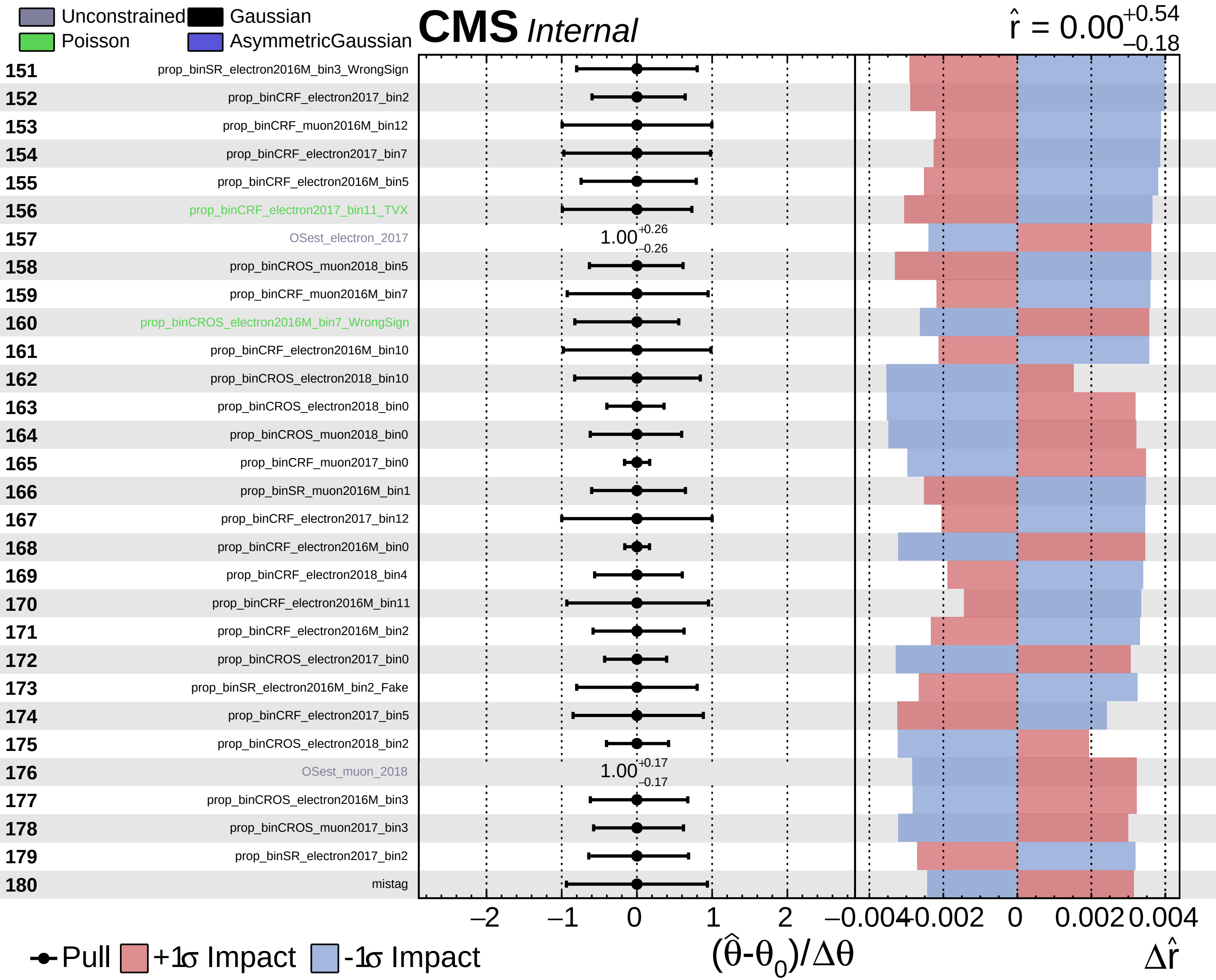
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
61 prop_binSR_muon2017_bin3
62 prop_binSR_electron2016M_bin6
63 QCDScale_TVX
64 prop_binSR_muon2016M_bin4
65 ISR_TVX
66 prop_binSR_muon2016M_bin3
67 prop_binCRF_muon2018_bin5
68 prop_binCRF_muon2018_bin9
69 ISR_Triboson
70 prop_binSR_electron2017_bin7
71 FSR_ZZtoLep
72 prop_binSR_muon2017_bin2
73 prop_binSR_electron2017_bin4
74 prop_binCRF_muon2018_bin10
75 prop_binSR_electron2016M_bin8
76 prop_binSR_electron2016M_bin5
77 prop_binCRF_muon2018_bin4
78 prop_binCRF_muon2018_bin11
79 PF
80 prop_binSR_muon2016M_bin2
81 prop_binSR_electron2016M_bin7
82 pdf_total_WpWpJJ_QCD
83 prop_binCRF_muon2018_bin8
84 prop_binCRF_muon2017_bin6
85 prop_binCRF_electron2018_bin6
86 prop_binCRF_muon2017_bin5
87 FSR_VG
88 prop_binSR_muon2017_bin1
89 prop_binCRF_electron2016M_bin9_WrongSign
90 prop_binCRF_muon2018_bin7
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

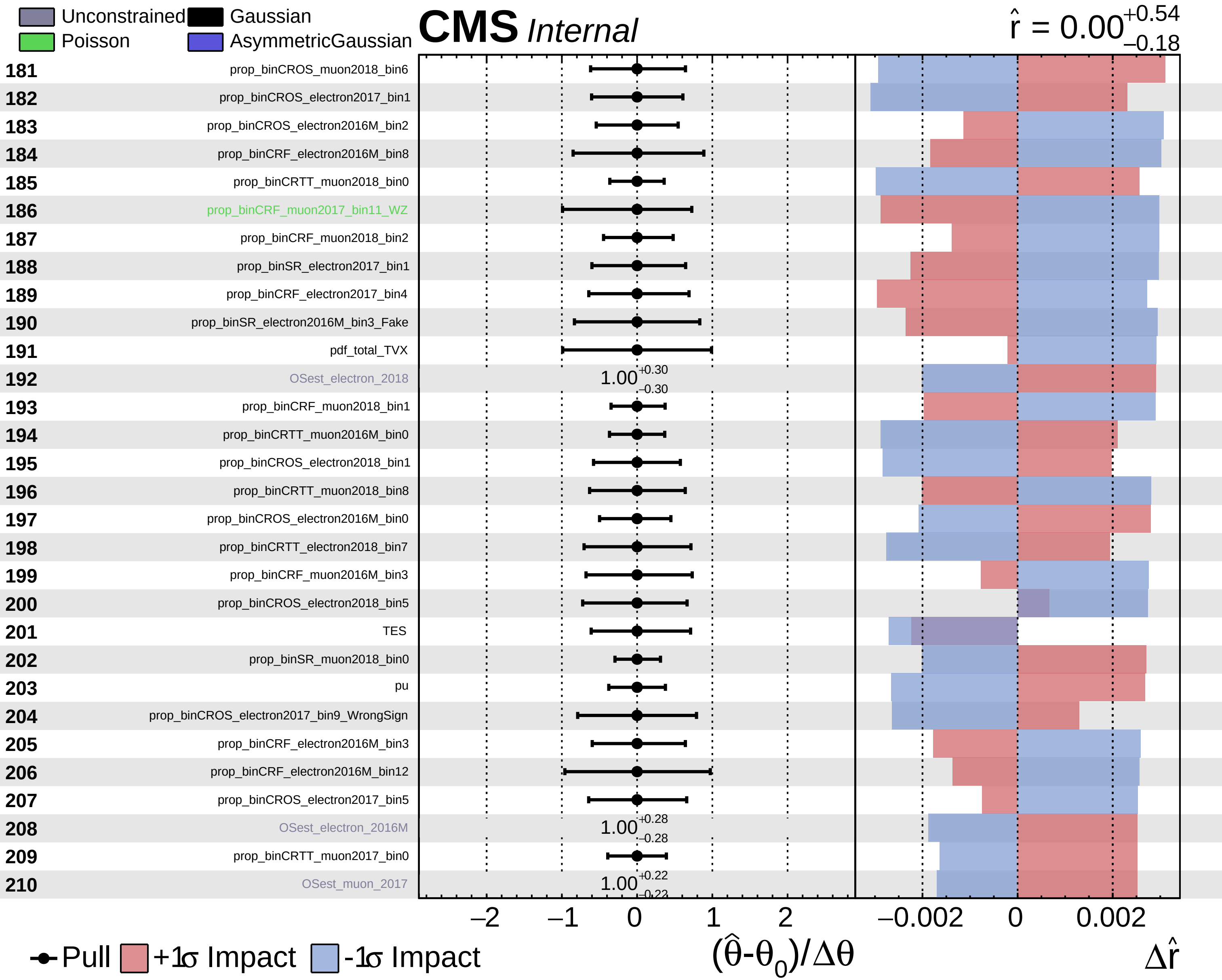


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
91 prop_binSR_electron2018_bin3
92 prop_binCROS_muon2016M_bin1
93 ISR_ZZtoLep
94 prop_binCRF_muon2017_bin10
95 prop_binCRF_electron2018_bin0
96 prop_binCRF_muon2017_bin4
97 prop_binCRF_electron2018_bin11
98 prop_binCRF_electron2018_bin10
99 prop_binCRF_electron2018_bin9
100 prop_binCRF_muon2016M_bin5
101 prop_binCROS_muon2016M_bin0
102 prop_binCRF_muon2017_bin3
103 prop_binCROS_electron2017_bin4
104 prop_binSR_electron2016M_bin4
105 btag
106 jes
107 prop_binCRF_muon2018_bin0
108 ISR_VG
109 prop_binCRF_muon2016M_bin6
110 prop_binSR_electron2017_bin3
111 prop_binCRF_muon2017_bin2
112 QCDScale_VG
113 prop_binCRF_muon2017_bin9
114 prop_binSR_muon2018_bin1
115 prop_binCRF_electron2017_bin0
116 prop_binCRF_electron2017_bin11_Fake
117 prop_binCRTT_electron2017_bin0
118 prop_binCRF_muon2017_bin8
119 prop_binCRF_electron2017_bin6
120 prop_binCROS_muon2018_bin3
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

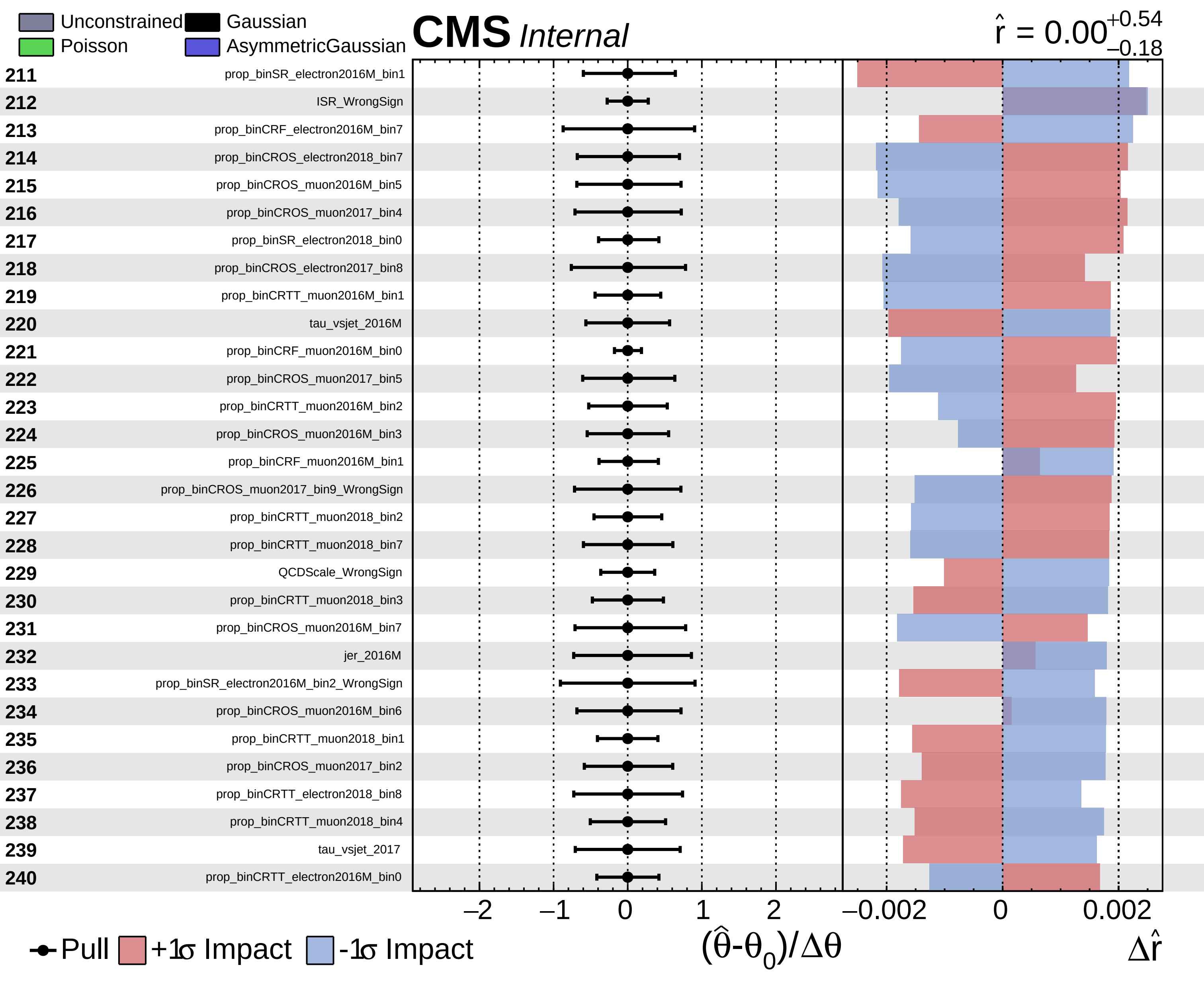


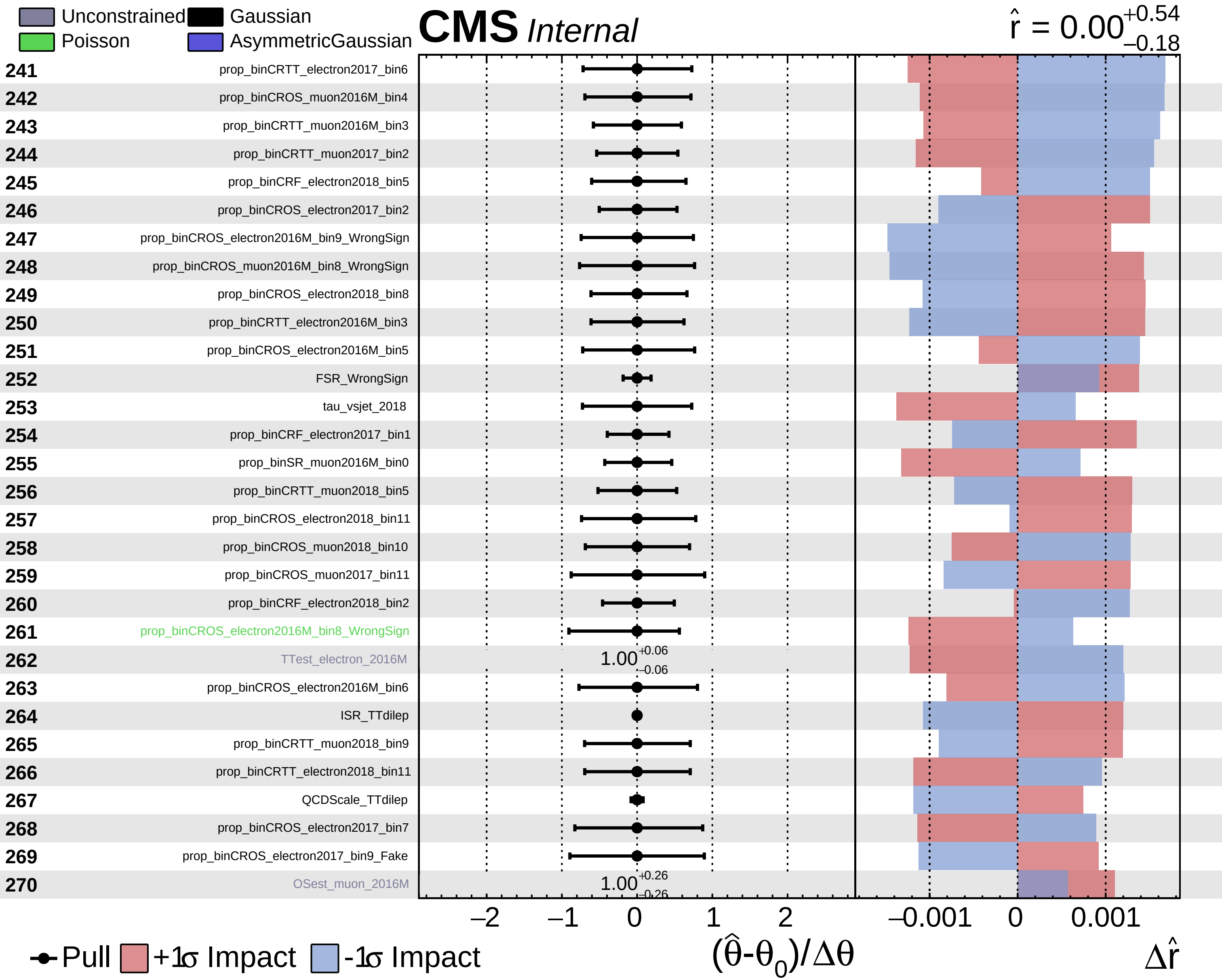
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
121 prop_binCRF_muon2017_bin11_Fake
122 prop_binCRF_muon2016M_bin4
123 prop_binCRF_electron2017_bin10
124 QCDScale_Triboson
125 prop_binSR_electron2018_bin2
126 prop_binCRF_muon2016M_bin9
127 prop_binCRF_electron2018_bin12
128 prop_binCRF_muon2016M_bin10
129 prop_binCRF_electron2018_bin3
130 prop_binCRF_electron2017_bin3
131 metUnclust
132 prop_binCRF_electron2018_bin7
133 prop_binCRF_muon2016M_bin11
134 prop_binCRF_muon2017_bin7
135 prop_binCRF_muon2018_bin3
136 prop_binCROS_muon2017_bin0
137 prop_binCRTT_electron2018_bin2
138 prop_binCRF_muon2018_bin12
139 prop_binCRF_electron2017_bin9
140 prop_binCRF_electron2018_bin8
141 prop_binCROS_muon2018_bin7
142 prop_binCRF_electron2017_bin11_TTTo2L2Nu
143 prop_binCRF_muon2016M_bin8
144 prop_binCRF_electron2016M_bin4
145 prop_binSR_electron2018_bin1
146 prop_binCROS_muon2018_bin2
147 prop_binCRF_muon2017_bin12
148 prop_binCRF_electron2016M_bin6
149 prop_binCROS_muon2018_bin8
150 prop_binCRF_electron2017_bin8
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.54 −0.18
151 prop_binSR_electron2016M_bin3_WrongSign
152 prop_binCRF_electron2017_bin2
153 prop_binCRF_muon2016M_bin12
154 prop_binCRF_electron2017_bin7
155 prop_binCRF_electron2016M_bin5
156 prop_binCRF_electron2017_bin11_TVX
157 OSest_electron_2017
1.00 +0.26 −0.26
158 prop_binCROS_muon2018_bin5
159 prop_binCRF_muon2016M_bin7
160 prop_binCROS_electron2016M_bin7_WrongSign
161 prop_binCRF_electron2016M_bin10
162 prop_binCROS_electron2018_bin10
163 prop_binCROS_electron2018_bin0
164 prop_binCROS_muon2018_bin0
165 prop_binCRF_muon2017_bin0
166 prop_binSR_muon2016M_bin1
167 prop_binCRF_electron2017_bin12
168 prop_binCRF_electron2016M_bin0
169 prop_binCRF_electron2018_bin4
170 prop_binCRF_electron2016M_bin11
171 prop_binCRF_electron2016M_bin2
172 prop_binCROS_electron2017_bin0
173 prop_binSR_electron2016M_bin2_Fake
174 prop_binCRF_electron2017_bin5
175 prop_binCROS_electron2018_bin2
176 OSest_muon_2018
1.00 +0.17 −0.17
177 prop_binCROS_electron2016M_bin3
178 prop_binCROS_muon2017_bin3
179 prop_binSR_electron2017_bin2
180 mistag
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

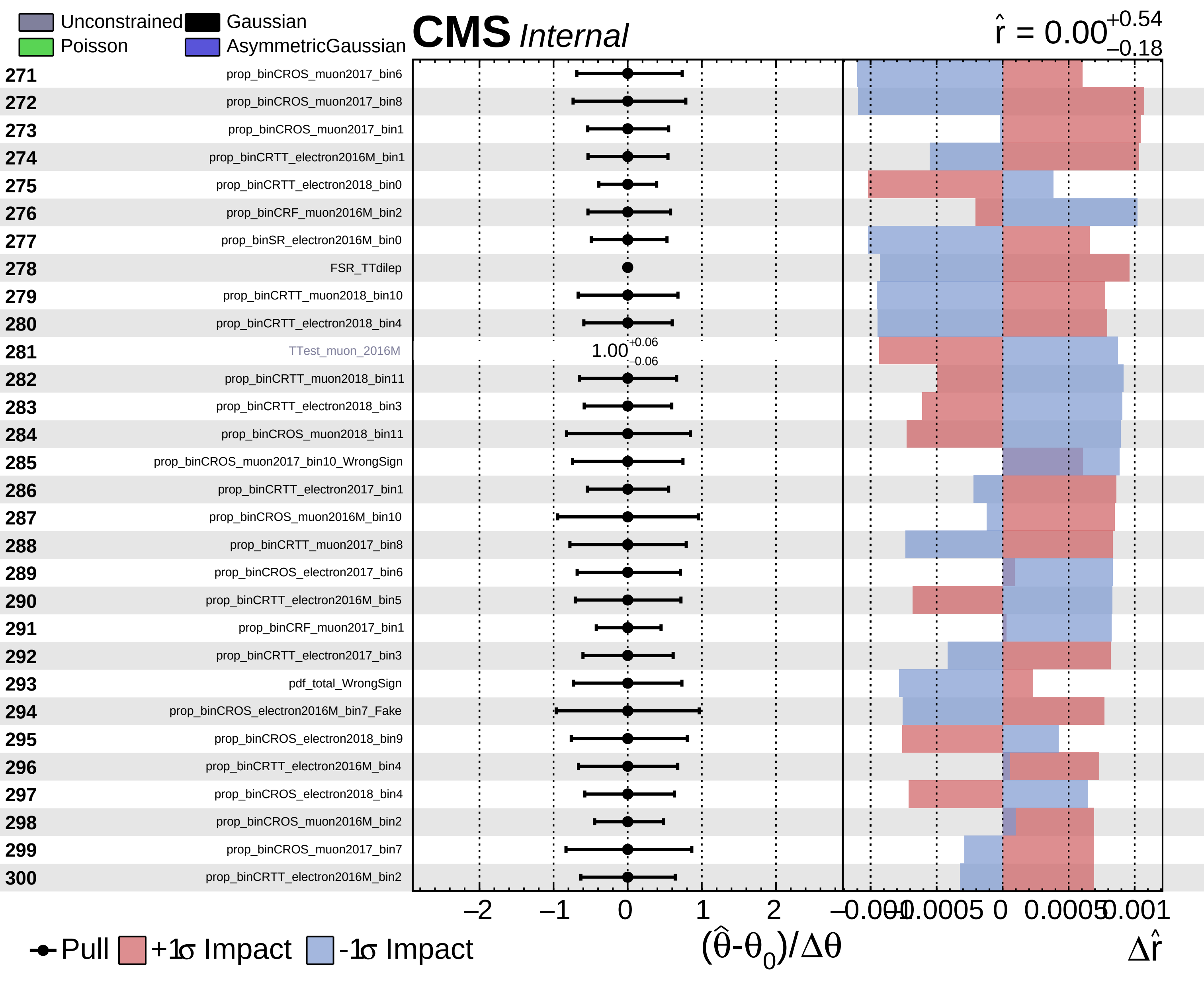


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
181 prop_binCROS_muon2018_bin6
182 prop_binCROS_electron2017_bin1
183 prop_binCROS_electron2016M_bin2
184 prop_binCRF_electron2016M_bin8
185 prop_binCRTT_muon2018_bin0
186 prop_binCRF_muon2017_bin11_WZ
187 prop_binCRF_muon2018_bin2
188 prop_binSR_electron2017_bin1
189 prop_binCRF_electron2017_bin4
190 prop_binSR_electron2016M_bin3_Fake
191 pdf_total_TVX
192 OSest_electron_2018
$1.00^{+0.30}_{-0.30}$
193 prop_binCRF_muon2018_bin1
194 prop_binCRTT_muon2016M_bin0
195 prop_binCROS_electron2018_bin1
196 prop_binCRTT_muon2018_bin8
197 prop_binCROS_electron2016M_bin0
198 prop_binCRTT_electron2018_bin7
199 prop_binCRF_muon2016M_bin3
200 prop_binCROS_electron2018_bin5
201 TES
202 prop_binSR_muon2018_bin0
203 pu
204 prop_binCROS_electron2017_bin9_WrongSign
205 prop_binCRF_electron2016M_bin3
206 prop_binCRF_electron2016M_bin12
207 prop_binCROS_electron2017_bin5
208 OSest_electron_2016M
$1.00^{+0.28}_{-0.28}$
209 prop_binCRTT_muon2017_bin0
210 OSest_muon_2017
$1.00^{+0.22}_{-0.22}$
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

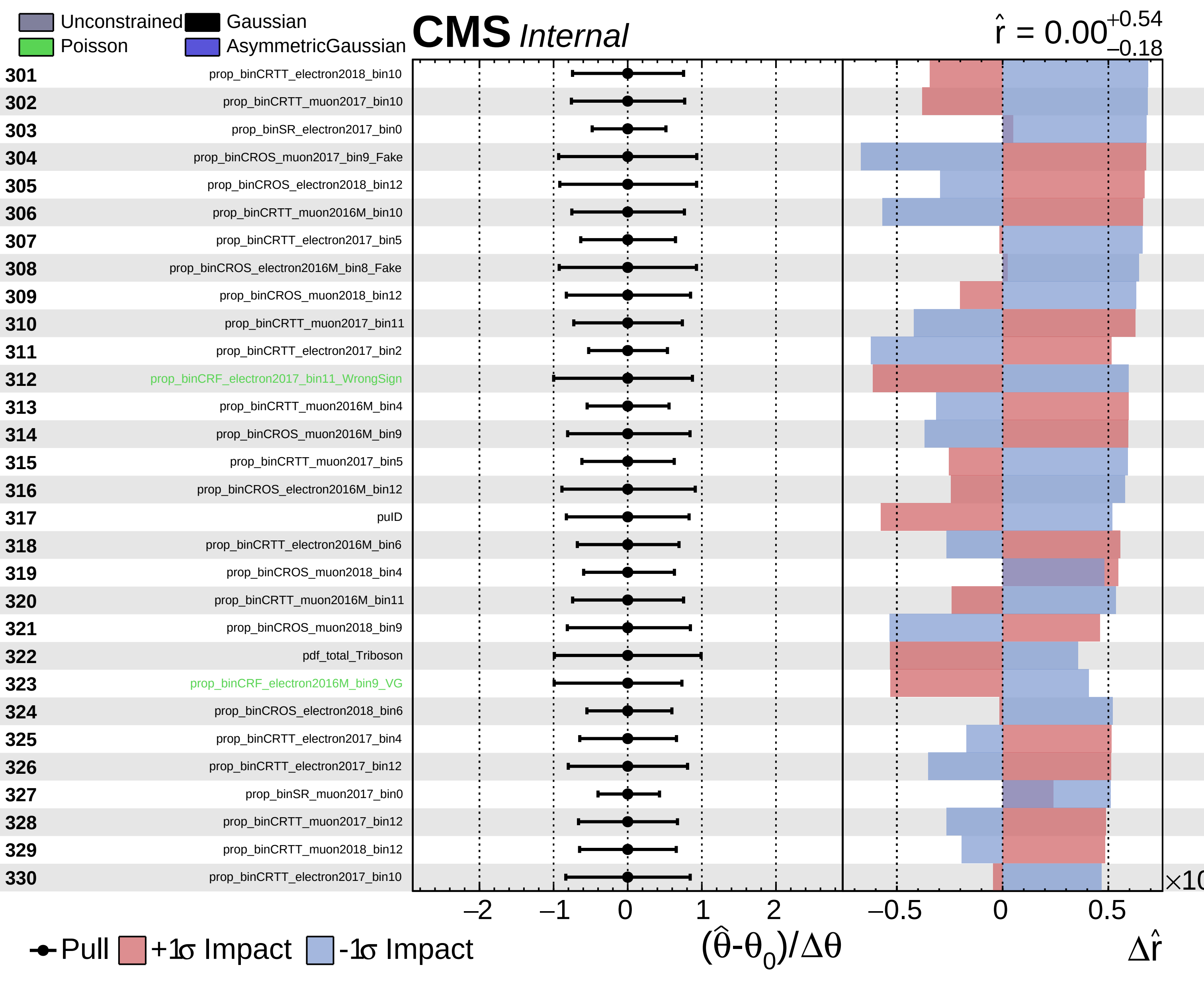

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
211 prop_binSR_electron2016M_bin1
212 ISR_WrongSign
213 prop_binCRF_electron2016M_bin7
214 prop_binCROS_electron2018_bin7
215 prop_binCROS_muon2016M_bin5
216 prop_binCROS_muon2017_bin4
217 prop_binSR_electron2018_bin0
218 prop_binCROS_electron2017_bin8
219 prop_binCRTT_muon2016M_bin1
220 tau_vsjet_2016M
221 prop_binCRF_muon2016M_bin0
222 prop_binCROS_muon2017_bin5
223 prop_binCRTT_muon2016M_bin2
224 prop_binCROS_muon2016M_bin3
225 prop_binCRF_muon2016M_bin1
226 prop_binCROS_muon2017_bin9_WrongSign
227 prop_binCRTT_muon2018_bin2
228 prop_binCRTT_muon2018_bin7
229 QCDScale_WrongSign
230 prop_binCRTT_muon2018_bin3
231 prop_binCROS_muon2016M_bin7
232 jer_2016M
233 prop_binSR_electron2016M_bin2_WrongSign
234 prop_binCROS_muon2016M_bin6
235 prop_binCRTT_muon2018_bin1
236 prop_binCROS_muon2017_bin2
237 prop_binCRTT_electron2018_bin8
238 prop_binCRTT_muon2018_bin4
239 tau_vsjet_2017
240 prop_binCRTT_electron2016M_bin0
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



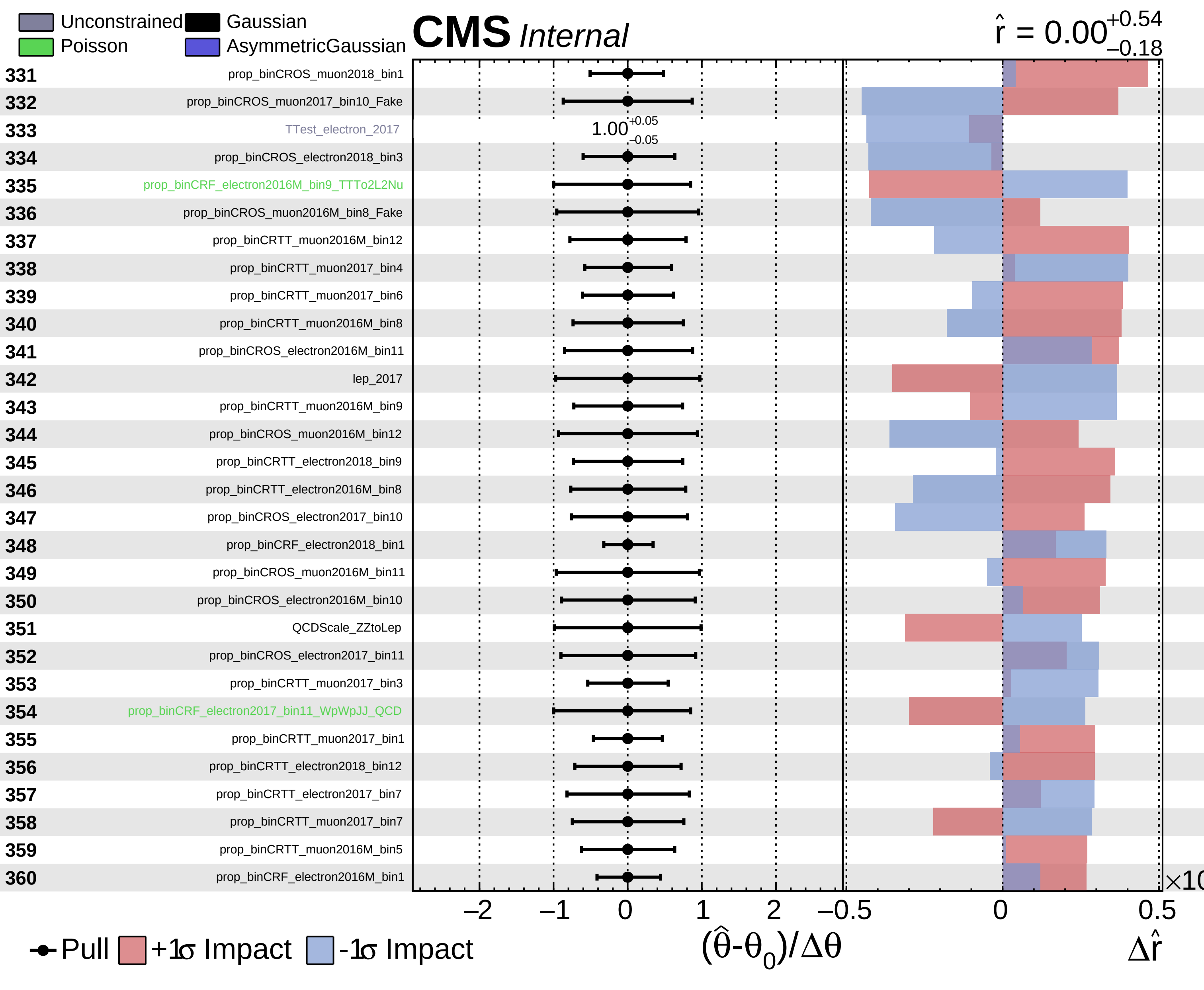
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
241 prop_binCRTT_electron2017_bin6
242 prop_binCROS_muon2016M_bin4
243 prop_binCRTT_muon2016M_bin3
244 prop_binCRTT_muon2017_bin2
245 prop_binCRF_electron2018_bin5
246 prop_binCROS_electron2017_bin2
247 prop_binCROS_electron2016M_bin9_WrongSign
248 prop_binCROS_muon2016M_bin8_WrongSign
249 prop_binCROS_electron2018_bin8
250 prop_binCRTT_electron2016M_bin3
251 prop_binCROS_electron2016M_bin5
252 FSR_WrongSign
253 tau_vsjet_2018
254 prop_binCRF_electron2017_bin1
255 prop_binSR_muon2016M_bin0
256 prop_binCRTT_muon2018_bin5
257 prop_binCROS_electron2018_bin11
258 prop_binCROS_muon2018_bin10
259 prop_binCROS_muon2017_bin11
260 prop_binCRF_electron2018_bin2
261 prop_binCROS_electron2016M_bin8_WrongSign
262 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
263 prop_binCROS_electron2016M_bin6
264 ISR_TTdilep
265 prop_binCRTT_muon2018_bin9
266 prop_binCRTT_electron2018_bin11
267 QCDScale_TTdilep
268 prop_binCROS_electron2017_bin7
269 prop_binCROS_electron2017_bin9_Fake
270 OSest_muon_2016M
$1.00^{+0.26}_{-0.26}$
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

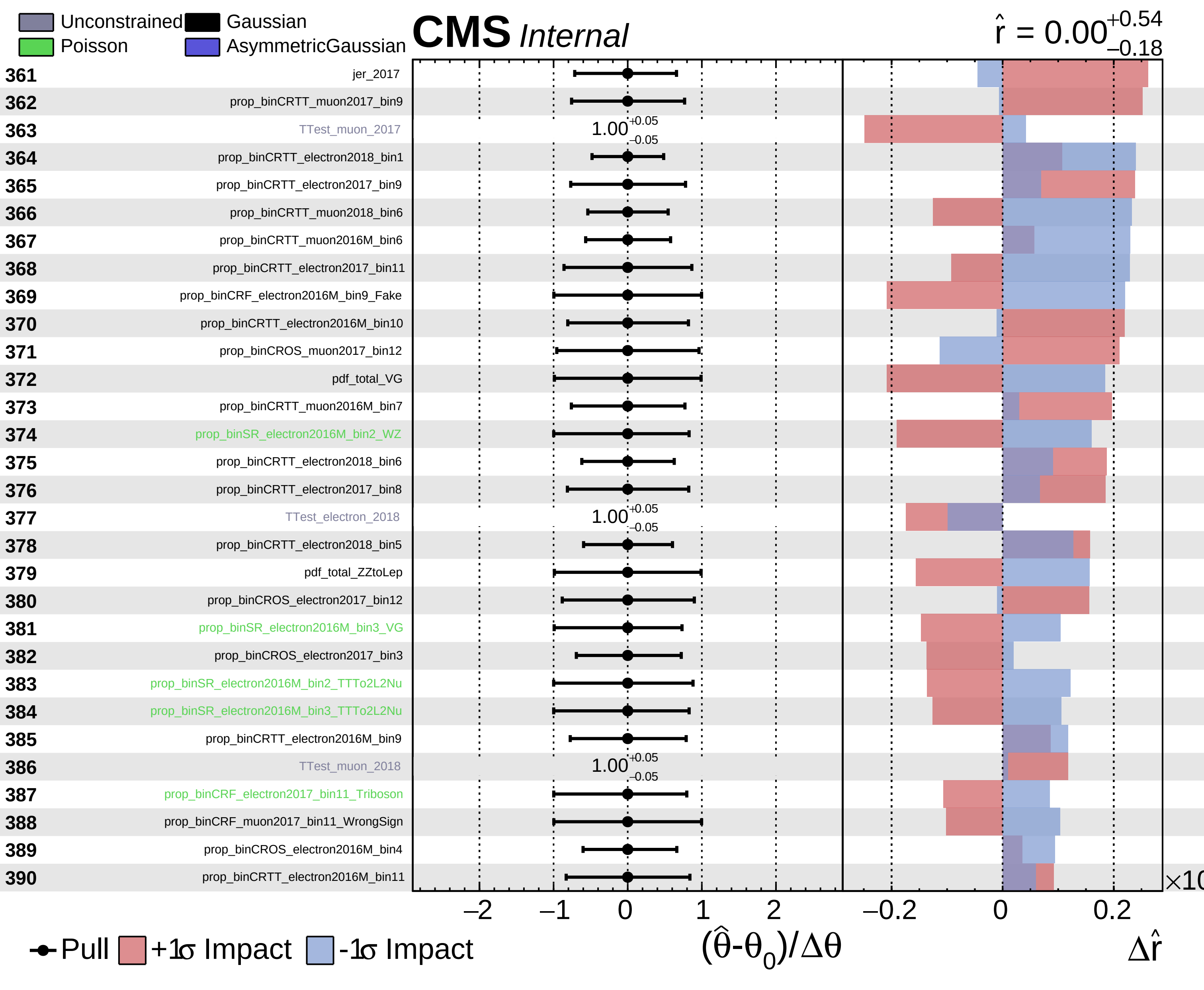


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
271 prop_binCROS_muon2017_bin6
272 prop_binCROS_muon2017_bin8
273 prop_binCROS_muon2017_bin1
274 prop_binCRTT_electron2016M_bin1
275 prop_binCRTT_electron2018_bin0
276 prop_binCRF_muon2016M_bin2
277 prop_binSR_electron2016M_bin0
278 FSR_TTdilep
279 prop_binCRTT_muon2018_bin10
280 prop_binCRTT_electron2018_bin4
281 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
282 prop_binCRTT_muon2018_bin11
283 prop_binCRTT_electron2018_bin3
284 prop_binCROS_muon2018_bin11
285 prop_binCROS_muon2017_bin10_WrongSign
286 prop_binCRTT_electron2017_bin1
287 prop_binCROS_muon2016M_bin10
288 prop_binCRTT_muon2017_bin8
289 prop_binCROS_electron2017_bin6
290 prop_binCRTT_electron2016M_bin5
291 prop_binCRF_muon2017_bin1
292 prop_binCRTT_electron2017_bin3
293 pdf_total_WrongSign
294 prop_binCROS_electron2016M_bin7_Fake
295 prop_binCROS_electron2018_bin9
296 prop_binCRTT_electron2016M_bin4
297 prop_binCROS_electron2018_bin4
298 prop_binCROS_muon2016M_bin2
299 prop_binCROS_muon2017_bin7
300 prop_binCRTT_electron2016M_bin2
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

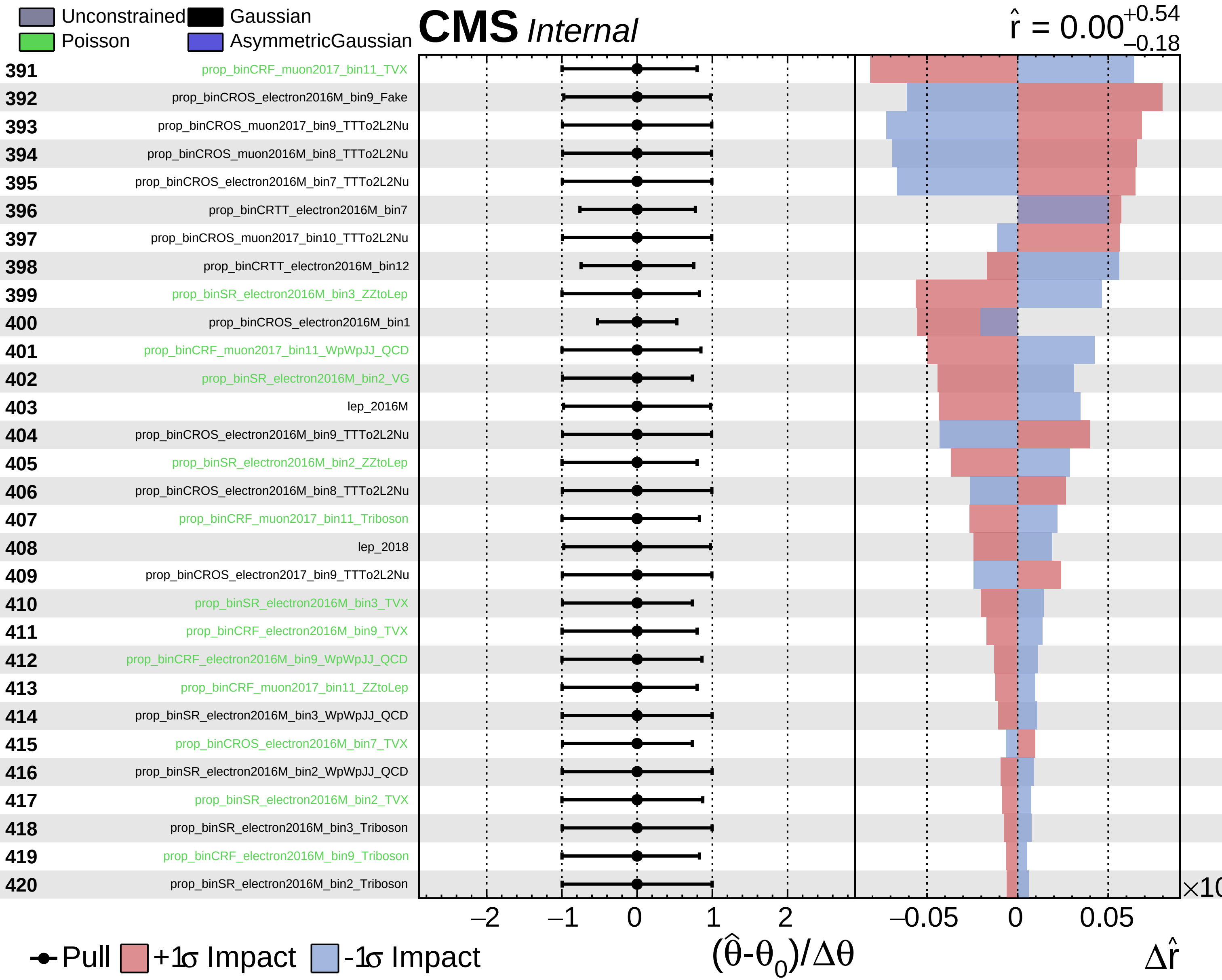


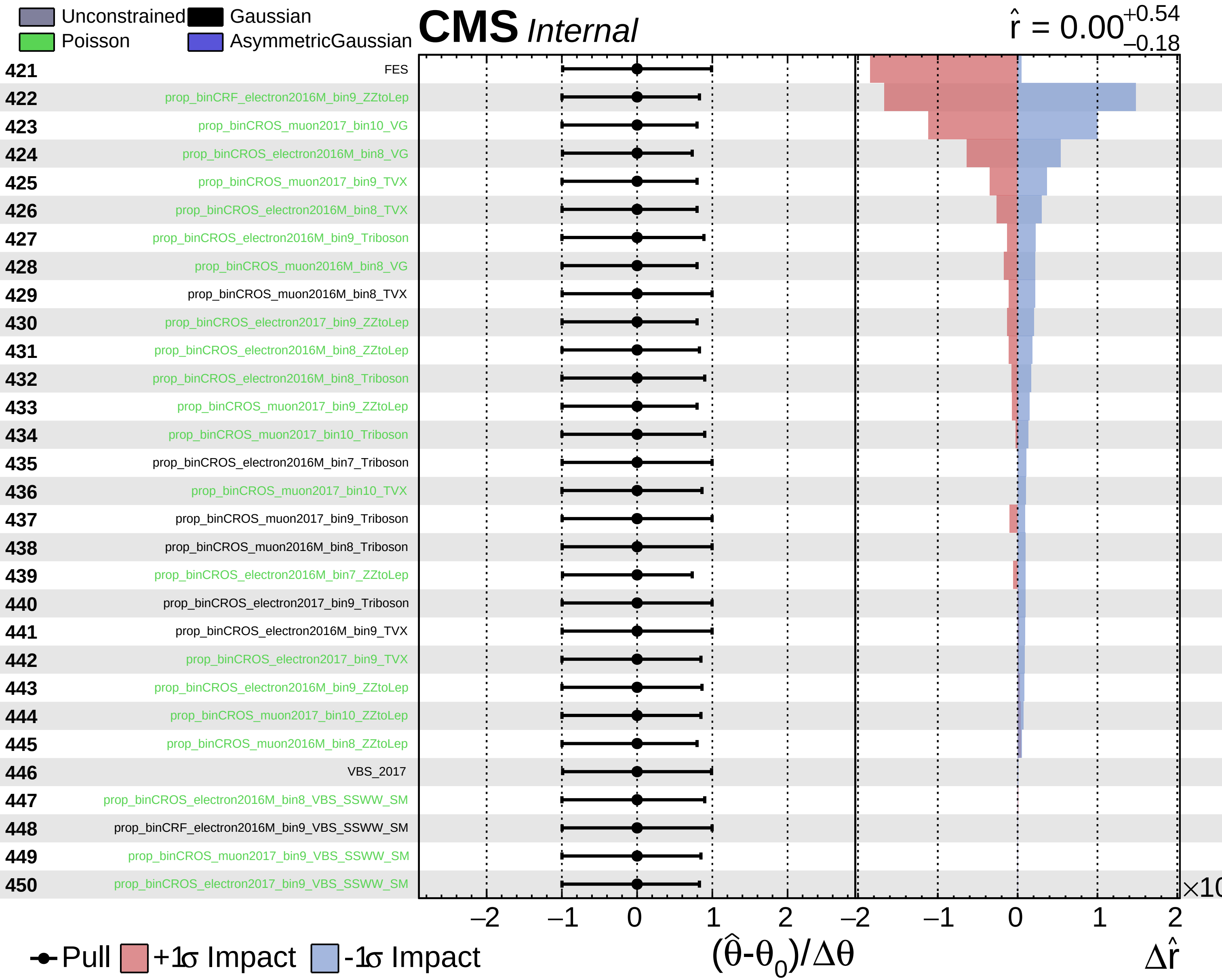
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
301 prop_binCRTT_electron2018_bin10
302 prop_binCRTT_muon2017_bin10
303 prop_binSR_electron2017_bin0
304 prop_binCROS_muon2017_bin9_Fake
305 prop_binCROS_electron2018_bin12
306 prop_binCRTT_muon2016M_bin10
307 prop_binCRTT_electron2017_bin5
308 prop_binCROS_electron2016M_bin8_Fake
309 prop_binCROS_muon2018_bin12
310 prop_binCRTT_muon2017_bin11
311 prop_binCRTT_electron2017_bin2
312 prop_binCRF_electron2017_bin11_WrongSign
313 prop_binCRTT_muon2016M_bin4
314 prop_binCROS_muon2016M_bin9
315 prop_binCRTT_muon2017_bin5
316 prop_binCROS_electron2016M_bin12
317 puID
318 prop_binCRTT_electron2016M_bin6
319 prop_binCROS_muon2018_bin4
320 prop_binCRTT_muon2016M_bin11
321 prop_binCROS_muon2018_bin9
322 pdf_total_Triboson
323 prop_binCRF_electron2016M_bin9_VG
324 prop_binCROS_electron2018_bin6
325 prop_binCRTT_electron2017_bin4
326 prop_binCRTT_electron2017_bin12
327 prop_binSR_muon2017_bin0
328 prop_binCRTT_muon2017_bin12
329 prop_binCRTT_muon2018_bin12
330 prop_binCRTT_electron2017_bin10
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
331 prop_binCROS_muon2018_bin1
332 prop_binCROS_muon2017_bin10_Fake
333 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
334 prop_binCROS_electron2018_bin3
335 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
336 prop_binCROS_muon2016M_bin8_Fake
337 prop_binCRTT_muon2016M_bin12
338 prop_binCRTT_muon2017_bin4
339 prop_binCRTT_muon2017_bin6
340 prop_binCRTT_muon2016M_bin8
341 prop_binCROS_electron2016M_bin11
342 lep_2017
343 prop_binCRTT_muon2016M_bin9
344 prop_binCROS_muon2016M_bin12
345 prop_binCRTT_electron2018_bin9
346 prop_binCRTT_electron2016M_bin8
347 prop_binCROS_electron2017_bin10
348 prop_binCRF_electron2018_bin1
349 prop_binCROS_muon2016M_bin11
350 prop_binCROS_electron2016M_bin10
351 QCDScale_ZZtoLep
352 prop_binCROS_electron2017_bin11
353 prop_binCRTT_muon2017_bin3
354 prop_binCRF_electron2017_bin11_WpWpJJ_QCD
355 prop_binCRTT_muon2017_bin1
356 prop_binCRTT_electron2018_bin12
357 prop_binCRTT_electron2017_bin7
358 prop_binCRTT_muon2017_bin7
359 prop_binCRTT_muon2016M_bin5
360 prop_binCRF_electron2016M_bin1
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

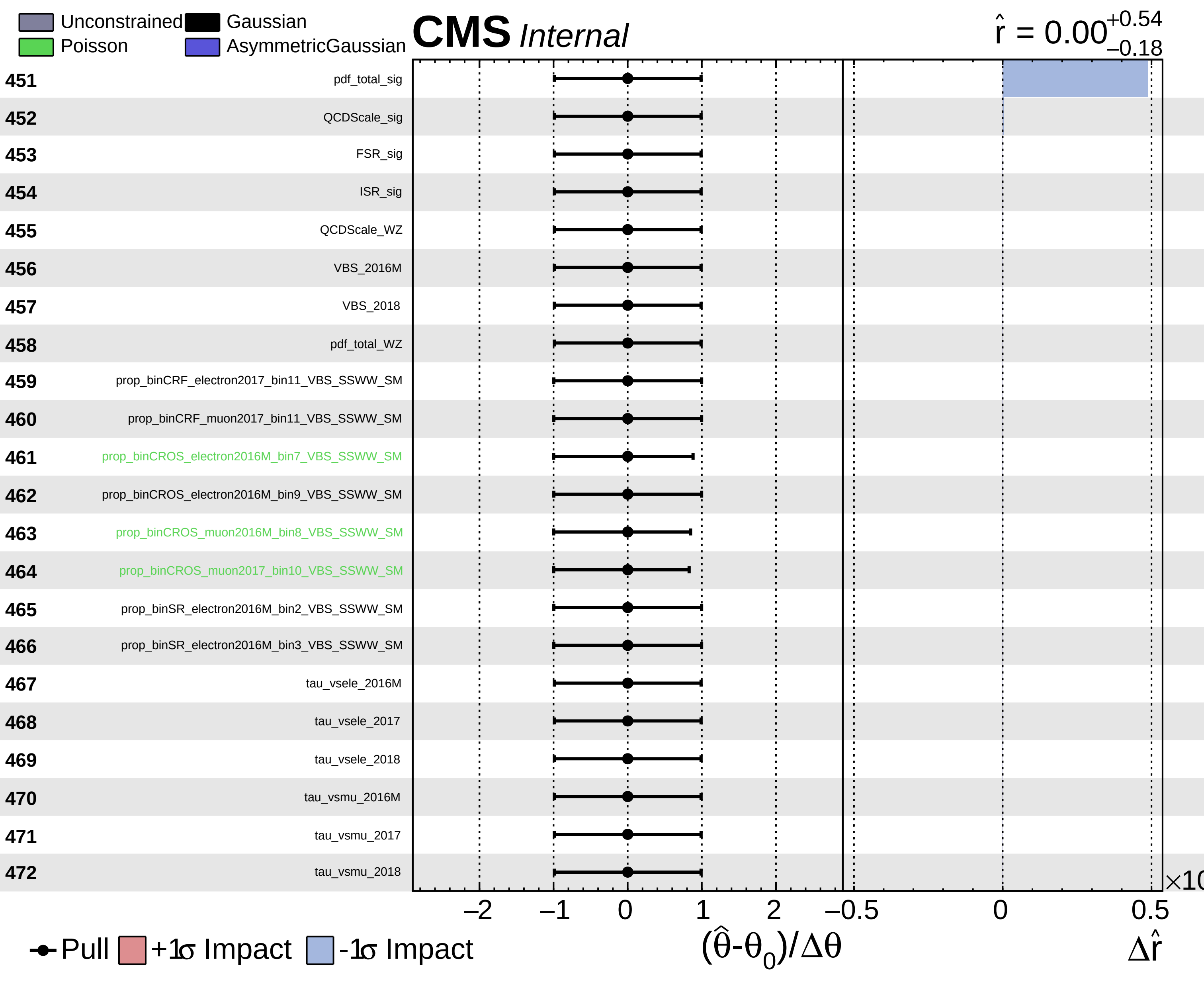


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
361 jer_2017
362 prop_binCRTT_muon2017_bin9
363 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
364 prop_binCRTT_electron2018_bin1
365 prop_binCRTT_electron2017_bin9
366 prop_binCRTT_muon2018_bin6
367 prop_binCRTT_muon2016M_bin6
368 prop_binCRTT_electron2017_bin11
369 prop_binCRF_electron2016M_bin9_Fake
370 prop_binCRTT_electron2016M_bin10
371 prop_binCROS_muon2017_bin12
372 pdf_total_VG
373 prop_binCRTT_muon2016M_bin7
374 prop_binSR_electron2016M_bin2_WZ
375 prop_binCRTT_electron2018_bin6
376 prop_binCRTT_electron2017_bin8
377 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
378 prop_binCRTT_electron2018_bin5
379 pdf_total_ZZtoLep
380 prop_binCROS_electron2017_bin12
381 prop_binSR_electron2016M_bin3_VG
382 prop_binCROS_electron2017_bin3
383 prop_binSR_electron2016M_bin2_TTTo2L2Nu
384 prop_binSR_electron2016M_bin3_TTTo2L2Nu
385 prop_binCRTT_electron2016M_bin9
386 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
387 prop_binCRF_electron2017_bin11_Triboson
388 prop_binCRF_muon2017_bin11_WrongSign
389 prop_binCROS_electron2016M_bin4
390 prop_binCRTT_electron2016M_bin11
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
391 prop_binCRF_muon2017_bin11_TVX
392 prop_binCROS_electron2016M_bin9_Fake
393 prop_binCROS_muon2017_bin9_TTTo2L2Nu
394 prop_binCROS_muon2016M_bin8_TTTo2L2Nu
395 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
396 prop_binCRTT_electron2016M_bin7
397 prop_binCROS_muon2017_bin10_TTTo2L2Nu
398 prop_binCRTT_electron2016M_bin12
399 prop_binSR_electron2016M_bin3_ZZtoLep
400 prop_binCROS_electron2016M_bin1
401 prop_binCRF_muon2017_bin11_WpWpJJ_QCD
402 prop_binSR_electron2016M_bin2_VG
403 lep_2016M
404 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
405 prop_binSR_electron2016M_bin2_ZZtoLep
406 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
407 prop_binCRF_muon2017_bin11_Triboson
408 lep_2018
409 prop_binCROS_electron2017_bin9_TTTo2L2Nu
410 prop_binSR_electron2016M_bin3_TVX
411 prop_binCRF_electron2016M_bin9_TVX
412 prop_binCRF_electron2016M_bin9_WpWpJJ_QCD
413 prop_binCRF_muon2017_bin11_ZZtoLep
414 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
415 prop_binCROS_electron2016M_bin7_TVX
416 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
417 prop_binSR_electron2016M_bin2_TVX
418 prop_binSR_electron2016M_bin3_Triboson
419 prop_binCRF_electron2016M_bin9_Triboson
420 prop_binSR_electron2016M_bin2_Triboson
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
421 FES
422 prop_binCRF_electron2016M_bin9_ZZtoLep
423 prop_binCROS_muon2017_bin10_VG
424 prop_binCROS_electron2016M_bin8_VG
425 prop_binCROS_muon2017_bin9_TVX
426 prop_binCROS_electron2016M_bin8_TVX
427 prop_binCROS_electron2016M_bin9_Triboson
428 prop_binCROS_muon2016M_bin8_VG
429 prop_binCROS_muon2016M_bin8_TVX
430 prop_binCROS_electron2017_bin9_ZZtoLep
431 prop_binCROS_electron2016M_bin8_ZZtoLep
432 prop_binCROS_electron2016M_bin8_Triboson
433 prop_binCROS_muon2017_bin9_ZZtoLep
434 prop_binCROS_muon2017_bin10_Triboson
435 prop_binCROS_electron2016M_bin7_Triboson
436 prop_binCROS_muon2017_bin10_TVX
437 prop_binCROS_muon2017_bin9_Triboson
438 prop_binCROS_muon2016M_bin8_Triboson
439 prop_binCROS_electron2016M_bin7_ZZtoLep
440 prop_binCROS_electron2017_bin9_Triboson
441 prop_binCROS_electron2016M_bin9_TVX
442 prop_binCROS_electron2017_bin9_TVX
443 prop_binCROS_electron2016M_bin9_ZZtoLep
444 prop_binCROS_muon2017_bin10_ZZtoLep
445 prop_binCROS_muon2016M_bin8_ZZtoLep
446 VBS_2017
447 prop_binCROS_electron2016M_bin8_VBS_SSWW_SM
448 prop_binCRF_electron2016M_bin9_VBS_SSWW_SM
449 prop_binCROS_muon2017_bin9_VBS_SSWW_SM
450 prop_binCROS_electron2017_bin9_VBS_SSWW_SM
−2 −1 0 1 2
−2 −1 0 1 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.54}_{-0.18}$
451 pdf_total_sig
452 QCDScale_sig
453 FSR_sig
454 ISR_sig
455 QCDScale_WZ
456 VBS_2016M
457 VBS_2018
458 pdf_total_WZ
459 prop_binCRF_electron2017_bin11_VBS_SSWW_SM
460 prop_binCRF_muon2017_bin11_VBS_SSWW_SM
461 prop_binCROS_electron2016M_bin7_VBS_SSWW_SM
462 prop_binCROS_electron2016M_bin9_VBS_SSWW_SM
463 prop_binCROS_muon2016M_bin8_VBS_SSWW_SM
464 prop_binCROS_muon2017_bin10_VBS_SSWW_SM
465 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
466 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
467 tau_vsele_2016M
468 tau_vsele_2017
469 tau_vsele_2018
470 tau_vsmu_2016M
471 tau_vsmu_2017
472 tau_vsmu_2018
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$